The USAPhO Exam

What it is, how it is run, and how to write answers that earn points

For official information, use the AAPT (American Association of Physics Teachers) website. Trust that website above anything in this handout.

www.aapt.org/physicsteam

The pages worth bookmarking are Rules, For Students, and Past Exams. Everything below comes from those pages or from the instruction pages of recent exams, but the exam changes — check before you plan around any of it.

1 The Short Version

The USAPhO is a three-hour, six-problem, free-response exam worth 150 points. It runs in two 90-minute halves of three problems each, with a break between, and you may not return to the first half. You qualify by clearing the cutoff on the F = ma exam in February; the USAPhO is in April. It is proctored in person. The questions are delivered on a computer, but **you write your solutions by hand on blank paper the proctor gives you**, and those pages are scanned and graded by humans. **You get a reference sheet** with constants and a few approximations. A basic scientific calculator is allowed. Almost nobody finishes, and partial credit is most of the game.

2 Part I: The Logistics

How do I qualify?

By scoring at or above the cutoff on the F = ma exam. AAPT sets that cutoff after the exam, at whatever score invites roughly 400–500 students; recent cutoffs have been 14–18 out of 25. There is no other route in — you cannot register for the USAPhO directly. See the *F = ma Exam FAQ* handout for everything about that exam.

When is it?

April, a few weeks after F = ma results come out. In 2026 it was **April 10**, inside a 1:00–5:00 pm EDT window (the exam is three hours plus a break; your proctor picks the start time inside the window). Watch AAPT's "Important Dates" page — in 2026 the date moved once, from April 2 to April 10, to avoid a Passover conflict.

What is the shape of the exam?

	Part A	**Part B**
Length	90 minutes	90 minutes
Problems	3 (A1, A2, A3)	3 (B1, B2, B3)
Points each	25	25
Part total	75	75

150 points total. Each problem has several lettered subparts. The instructions say plainly that problems worth the same points "may differ in difficulty," so the order on the page is *not* a difficulty ranking.

Part A and Part B are sealed off from each other. You may not look at Part B during Part A, and after the break you may not go back to Part A — your Part A pages have already been collected. Treat them as two separate 90-minute exams. Whatever you did not finish in Part A is gone.

What topics does it cover?

All of introductory physics: mechanics, electricity and magnetism, thermodynamics and statistical physics, waves, optics, and relativity and modern physics. Unlike F = ma, it is **not** mechanics-only, and unlike F = ma, **calculus is expected.** You will integrate, differentiate, expand in small parameters, and solve differential equations. The 2026 exam included a geometrical optics problem, a problem about motion on a cone, a blackbody radiation problem, and a rigid-body problem, among others.

Is it on paper or on a computer?

Both, and this is the part people get wrong. Since the 2026 cycle AAPT states that the F = ma and USAPhO exams are "conducted online via Educational Vistas only," with "no printable versions of the exams available." What that means in practice, from the 2025 and 2026 student instructions:

- You sit at a designated desktop, laptop, or Chromebook. Phones, tablets, and smartwatches are handed in before the exam starts.
- **You read the problems on the screen.** When the exam begins you click the link for Part A and a 90-minute timer starts.
- **You write your solutions by hand**, on blank paper the proctor hands you, along with a printed instruction page, the reference sheet, and a cover sheet for each problem.
- At the end of each part the proctor collects your written pages and scans them. Those scans are what get graded.

The delivery is online; the answering is not. You are still writing physics on paper with a pen, and a human is still reading it.

Do I get scratch paper, and how should I use it?

Yes, from the proctor only — "Only scratch paper provided by the proctors may be used" — and it is all collected at the end. You take none of it home.

The rule that matters is the one about what gets *graded*: "All work must be done on the provided blank pages... Work outside of these pages will not be graded," and when the proctor collects your solutions, "Do not include scratch work." **Scratch paper is invisible to the grader.**

Use scratch paper for deciding, not for solving. Scratch is the right place to try three approaches and see which two collapse. Once you know which one works, **switch to the answer pages and do the real derivation there.** Do not solve the whole problem on scratch and then copy it over — that is doing the problem twice, and you do not have time to do six problems twice. Grinding through a long integral on scratch and transcribing only the final line is the worst of both worlds: you spent the time and the grader cannot see the work you spent it on.

Is there a formula sheet?

Yes — a real difference from F = ma, which gives you nothing. The USAPhO reference sheet is printed in the instructions and you may use it for both parts. In 2026 it held:

- **Constants:** g, G, $k = 1/4\pi\epsilon_0$, $k_m = \mu_0/4\pi$, c, k_B, N_A, R, σ, e, 1 eV in joules, h, and m_e.
- **Approximations:** $(1+x)^n \approx 1 + nx + \frac{1}{2}n(n-1)x^2$, $e^x \approx 1 + x + x^2/2 + x^3/6$, $\sin\theta \approx \theta - \theta^3/6$, and $\cos\theta \approx 1 - \theta^2/2$.

That is the whole sheet. **It is a constants sheet, not a formula sheet.** There is no $\frac{1}{2}mv^2$, no Maxwell's equations, no moments of inertia, no thermodynamic relations, no lens equation. Some years have also carried a short table of integrals and trigonometric sum identities; the 2026 sheet did not. Do not count on it. Know your formulas.

That they hand you those four approximations is a hint about the exam's taste: **expansions in a small parameter show up constantly**. When a problem hands you a condition like $r \ll R$, it is telling you which expansion to run.

What are the calculator rules?

A handheld calculator is allowed, but it "may not be a graphical calculator, its display has no more than three lines," and its memory must be cleared of data and programs before the exam. No symbolic math, no programming, no graphing. Calculators may not be shared. Bring your own; the proctor will not have one. You will use it much less than on F = ma — most answers are symbolic.

What else is banned?

Books, tables, collections of formulas, notes — anything beyond the provided reference sheet. Phones and other internet-connected devices are handed in and may not be used at any time while exam materials are present. You must stay in the exam room for the entire testing period.

How is it proctored?

In person and live, like F = ma. Proctors need at least a two-year degree, need not be physics teachers, and **may not be a parent, relative, or close friend** of a student taking the exam. For-profit test prep centers are not allowed to proctor. If you qualify, AAPT emails you and your proctor, and usually the same teacher who ran your F = ma administration runs this one.

Can I talk about the problems afterward?

Not until the embargo lifts, which is typically the day after the exam — the 2025 exam told students not to discuss any part of it until April 11, the day after. Violations can mean disqualification. Solutions are posted publicly a few weeks later.

What happens after?

Results and an honors list come out in May. About 20 students, chosen on combined USAPhO and F = ma performance, are named to the **U.S. Physics Team** and invited to a ten-day training camp at the University of Maryland in late May. AAPT's standing description of what happens next is that "at the end of the camp, five students will be selected to represent the team at the International Physics Olympiad" — those five come out of the exams given during the camp itself.

> **2026 did it differently, and it is not yet clear whether that sticks.** A new exam, the **USAPhO Plus**, was given on May 10 to the 20 students *already* named to the team, and it — not the camp — chose the five travelers and one alternate. The camp, renamed a Training & Development Camp, ran afterward (May 30–June 9) to prepare that delegation and to develop future competitors. The 2026 team also competed at the **European Physics Olympiad** in Gothenburg rather than the IPhO.
> There was no USAPhO Plus in 2024 or 2025; in 2025 the five IPhO travelers were named after camp in the usual way. **None of this changes anything you do on exam day** — but if you are in range of the top 20, read AAPT's news page for your year rather than assuming either pattern.

One feature of the 2026 USAPhO Plus is worth knowing about regardless: alongside the usual theory problems it included a problem with an **experimental component** — analyzing experimental-style data, constructing and interpreting plots, extracting physical parameters, and evaluating uncertainties. Data analysis is also listed in the scope of the USAPhO itself, and it is the part of the syllabus American students most often skip.

3 Part II: How to Write an Answer

Start by picturing the grader

Your solution is scanned and read by a person who has already read a few hundred versions of the same problem, working from a rubric that assigns points to specific milestones: the right principle, the right setup, the right intermediate expression, the right final answer. They are not trying to reconstruct your thinking. They are trying to find the milestones.

Everything below follows from that one fact.

> **The five habits.**
>
> 1. **Draw a labeled diagram and define your symbols on it.** Every angle, distance, and force you use should be visible somewhere. A grader who cannot tell what your ℓ is cannot give you credit for an equation containing ℓ.

2. **Name the principle before you use it.** One short line — "torque balance about the contact point in the rotating frame," "energy conservation between release and the bottom," "Gauss's law on a cylinder of radius r." This is usually worth points on its own, and it costs eight seconds.
3. **Work symbolically; substitute numbers last.** Symbols show the grader the structure. A page of decimals shows them nothing, and one arithmetic slip poisons everything downstream.
4. **Box the final answer of every subpart.** The instructions explicitly ask for this: "To help with grading, please draw a clear box around your final answer to each subpart." An unboxed answer buried mid-page gets missed.
5. **One problem per sheet, header on every page.** Never mix two problems on one sheet. Label every page in the upper right.

What the page looks like

The exact answer-sheet format has drifted — for years AAPT printed framed answer sheets with an information block, and more recently you get a cover sheet per problem plus blank paper. What has not changed is the header AAPT asks for in the upper right corner of every page:

Student AAPT ID #	Proctor AAPT ID #
A1 — 1/3	

That last line is the one people forget and the one that matters most: **problem number, then page n of N for that problem**. The scans get split apart and sorted by problem before grading. A page labeled only "2" can end up in the wrong pile, and a page in the wrong pile scores zero.

Practical rules that follow from the pages being scanned and sorted:

- **Write on one side only.** Older instructions said so outright; graders scan fronts.
- **Number every page as n/N and fill in N at the end**, when you know it.
- **Press hard and write dark.** Faint pencil scans badly. Pen is safer, which is why crossing out beats erasing.
- **Leave the margins alone.** Anything in the outer half inch may be cropped.
- If you continue a problem on extra pages, **say so at the bottom of the previous page** ("continued on A1 – 3/3").

What earns partial credit, and what earns nothing

Earns points	Earns nothing
A named principle, even with no algebra after it	A page of algebra with no statement of what is being computed
A correct setup with the integral left unevaluated	A correct final answer with no work
A labeled diagram defining your variables	Symbols that appear only in equations, never defined
"I could not do (a); assume the answer is X" and then a correct (b)	A blank (b) because (a) did not work out
One clearly boxed answer per subpart	Three candidate answers, none crossed out
A wrong answer whose method is visible and mostly right	A wrong answer whose method is invisible

Never leave two live answers on the page. If you tried something, abandoned it, and tried again, put one clean line through the dead attempt. Graders are generally instructed to grade what you present, and if you present two contradictory answers they are entitled to grade the wrong one. Crossing out is free. Do it.

Getting past a part you cannot do

This is the single most valuable habit on this exam, and it is the one students most often fail to use.

Subparts are usually graded independently. If part (b) says "using your result from part (a)," you do not need part (a) to be *right* — you need it to be *stated*. So:

When you are stuck on (a) and (b) depends on it, write:

"Could not complete (a). For (b) I assume the result of (a) is $T = \ldots$."

and then do (b) properly. Full marks on (b) are still available. Two more versions of the same move:

- **Carry a symbol you cannot evaluate.** If you cannot find the moment of inertia, call it I, solve the rest of the problem in terms of I, and box that. You will lose the points for I and keep the rest.
- **Do the special case.** If the general problem is beyond you, solve it for the symmetric case, or to leading order in the small parameter, and say clearly that this is what you did. It is worth far more than a blank.

Two more things that reliably earn points

Check limiting cases, out loud, on the page. When you get an answer, test it: does it reduce to the earlier part when the correction goes to zero? Is the sign right? Do the units work? Writing "as $h \to 0$ this reduces to part (a) ✓" does two jobs: it catches your own errors, and it shows a grader who is deciding between two rubric levels that you understand what you produced.

Say what you are about to do before you do it. Two-word headers — "Setup," "Torque about P," "Evaluate," "Check" — turn a wall of algebra into a readable argument. On a page that gets four minutes of attention, structure is worth points.

4 Part III: Strategy for the Three Hours

The first five minutes: orient, do not solve

Do not start problem A1 the moment the clock starts. Spend the first three to five minutes reading all three problems in the part. You are looking for:

- Which topic each problem is (mechanics? thermo? circuits?).
- How many subparts each has, and whether the early subparts look approachable.
- Which problem is *yours* — the one in your strongest area.

This costs five minutes out of ninety and it protects you from the worst outcome on this exam: spending forty minutes on the hardest problem of the set because it happened to be printed first, and never reading a problem you could have solved.

The order on the page is not the order of difficulty. AAPT says so explicitly. A3 is often easier than A2.

Problem by problem, not skipping around

Once you have chosen an order, **work one problem at a time, start to finish.** Long free-response physics is not like multiple choice: there is a large cost to reloading a problem's setup into your head, so ping-ponging between problems wastes several minutes each time.

Within a problem, though, skip freely. If part (b) is stuck, read (c) and (d) immediately — they are often independent, or they depend on (b) in a way you can bridge with an assumed result. Some of the cheapest points on the exam sit in the last subpart of a problem whose middle nobody finished.

A time budget

Ninety minutes, three problems: **30 minutes each**, and the discipline is in the checkpoints, not the plan.

Clock	Where you should be
0:00–0:05	Read all three problems. Pick an order.
0:05–0:32	First problem (your best topic).
0:32–0:59	Second problem.
0:59–1:26	Third problem.
1:26–1:30	Harvest: box unboxed answers, fix headers, cross out dead work.

Hard stop at 30 minutes per problem, even mid-derivation. Write down where you got to, box whatever partial result you have, and move on. If you have time at the end you can come back — and you will come back with a fresh head, which is often what you needed. The alternative, sinking 50 minutes into one problem, costs you the first easy subparts of a problem you never read.
The five-minute rule for a stuck subpart: if five minutes of real effort produces no new equation, that subpart is over. Write what you know, state your assumption, and go to the next subpart.

The last four minutes of each part

Stop doing physics and go harvest points:

1. Box every final answer that is not boxed.
2. Put the problem number and n/N on every page; fill in N.
3. Cross out every abandoned attempt so nothing contradictory is left live.
4. For every subpart you left blank, write *one line* — the principle you would have used, the diagram you would have drawn, the form you expect the answer to take. A sentence beats a blank, and sometimes it is a rubric line.
5. Sort your pages by problem, then by page number.

Calibrate: you are not going to get 150

This matters, because the biggest self-inflicted wound on this exam is a student who treats an unfinished problem as a failure and spirals. The USAPhO is built so that the top students in the country do not finish it.

Solving two problems completely and making real progress on two more is an excellent performance. A blank subpart is normal. Three blank subparts is normal. **Your job is not to finish the exam — it is to collect the most points available to you in 180 minutes.** Those are different goals and they lead to different behavior: the second one makes you leave a hard problem, read the next one, and pick up its first two subparts.

5 Part IV: What Score Do I Need?

AAPT does not publish USAPhO score cutoffs, and has said it deliberately does not release numerical scores or detailed breakdowns. What it does publish is the *structure* of the awards, which is enough to aim by. Awards are given by percentile of the students who take the exam:

Award	**Share of takers**	**2026: about**
Gold Medal	top 10–12%	top 38 students
Silver Medal	next 14–16%	next 48
Bronze Medal	next 20–22%	next 63
Honorable Mention	next 24–26%	next 79
Distinction	remainder	the rest

Above all of that sits the **U.S. Physics Team**: about 20 students invited to camp, on combined USAPhO and F = ma performance.

2026 numbers: 440 students qualified on F = ma and 317 actually sat the USAPhO.

Two things fall out of that table. First, **a medal of some color goes to roughly the top 47% of the field**, and some honor goes to nearly three quarters of it. Second, **the field is 300-odd people who already finished in the top 7% of F = ma takers.** Middle of this pack is a genuinely strong physics student.

A rough way to think about targets. One problem is 25 points, so on the 150-point scale, in a typical year:

- **Honorable mention** is in the neighborhood of one problem's worth of credit — which in practice means solid partial credit spread over four or five problems, not one perfect problem and five blanks.
- **Bronze** is around one and a half problems' worth; **silver** around two; **gold** around two and a half or more.

These are rules of thumb for calibration, not published cutoffs, and they move year to year with the difficulty of the exam. The lesson is the same either way: **scattered partial credit across all six problems outscores heroic effort on one.**

6 Part V: What a Real Submission Looks Like

Everything above is abstract until you see a page. So here is one problem worked the way you would actually work it under time pressure — including a wrong turn — followed by how it would be graded.

The problem is **2014 USAPhO, Problem A1**, which is a good one to practice on: two subparts, the second a genuine correction to the first, and a trap in the middle that most people fall into.

Question A1 (2014 USAPhO). A unicyclist of total height h goes around a circular track of radius R while leaning inward at an angle θ to the vertical. The acceleration due to gravity is g.

(a) Suppose $h \ll R$. What angular velocity ω must the unicyclist sustain?

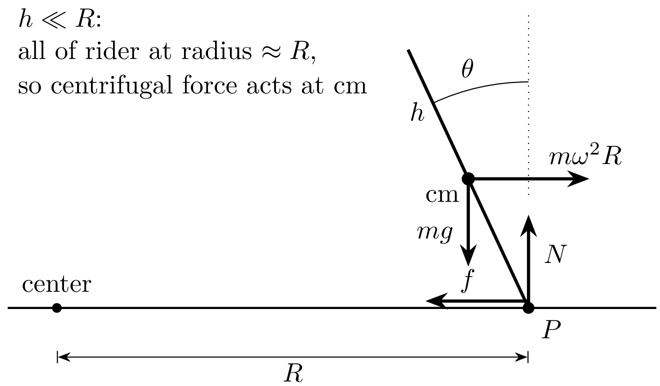
(b) Now model the unicyclist as a uniform rod of length h, where h is less than R but not negligible. This refined model introduces a correction to the previous result. What is the new expression for the angular velocity ω? Assume that the rod remains in the plane formed by the vertical and radial directions, and that R is measured from the center of the circle to the point of contact at the ground.

The submission

What follows is typeset so you can read it, but the content, the length, and the mess are what a good submission actually looks like. The notes in red are mine, pointing out the choices that are earning points; they would not be on your page.

Student AAPT ID # Proctor AAPT ID #
A1 — 1/2

(a) Rotating frame, rider static. Radial-vertical plane:



↑ The diagram is the setup. Every symbol used below — P, θ, cm, R — is defined on it, and the choice of frame is stated in the first line.

Let cm be a distance ℓ up the rod from P. Torque about P (rider is static in this frame, so $\tau_{\text{net}} = 0$; N and f act at P and contribute nothing):

$$\tau = m\omega^2 R\,(\ell\cos\theta) \;-\; mg\,(\ell\sin\theta) = 0$$

↑ "Torque about P" named, and the reason N and f drop out given in half a line. This is a rubric line.

$$\omega^2 = \frac{g\tan\theta}{R}$$

~~$\omega^2 = g\cot\theta/R$ — lever arms swapped~~ ← First attempt, caught and crossed out. See the check below.

$$\boxed{\omega = \sqrt{\frac{g\tan\theta}{R}}}$$

Check: $\theta \to 0$ (upright) $\Rightarrow \omega \to 0$. ✓ $\theta \to 90^\circ \Rightarrow \omega \to \infty$. ✓ Units: $\sqrt{(\mathrm{m/s^2})/\mathrm{m}} = \mathrm{s^{-1}}$. ✓
↑ This check is what caught the swapped lever arms. Three lines, thirty seconds, and it is the difference between 8/8 and 3/8 on this part.

Student AAPT ID # Proctor AAPT ID #
A1 — 2/2

(b) Now $h \not\ll R$, so different parts of the rod are at different radii and the centrifugal force is *not* uniform.

~~Idea: cm sits at radius $R - \frac{h}{2}\sin\theta$, so replace $R \to R - \frac{h}{2}\sin\theta$ in (a).~~

← The trap, tried and abandoned. Note the one-line reason written next to it rather than a silent erasure:

No — torque involves the product $r\,z$, and $\langle rz\rangle \neq \langle r\rangle\langle z\rangle$. Must integrate.

Setup. Mass element a distance s along the rod from P:

$$r = R - s\sin\theta, \qquad z = s\cos\theta, \qquad dm = \frac{m}{h}\,ds, \qquad 0 \le s \le h$$

↑ Each of the three is a rubric line on its own. Even if the integral had gone nowhere, these are already worth points.

Centrifugal torque about *P*. Force $\omega^2 r\,dm$ outward, lever arm z:

$$\tau_c = \int_0^h \omega^2 (R - s\sin\theta)(s\cos\theta)\,\frac{m}{h}\,ds$$

$$= \frac{m\omega^2\cos\theta}{h}\left[\frac{Rh^2}{2} - \frac{h^3\sin\theta}{3}\right] = m\omega^2 h\cos\theta\left(\frac{R}{2} - \frac{h\sin\theta}{3}\right)$$

Gravity torque about *P*, at the cm, $h/2$ up the rod:

$$\tau_g = -\,mg\,\frac{h}{2}\sin\theta$$

Balance. $\tau_c + \tau_g = 0$:

$$\omega^2 h\cos\theta\left(\frac{R}{2} - \frac{h\sin\theta}{3}\right) = g\,\frac{h}{2}\sin\theta \quad\Longrightarrow\quad \omega^2 = \frac{g\tan\theta}{R - \frac{2h}{3}\sin\theta}$$

$$\boxed{\omega = \sqrt{\frac{g}{R}\tan\theta\left(1 - \frac{2h}{3R}\sin\theta\right)^{-1}}}$$

Checks: $h \to 0$ reduces to (a) ✓ Denominator $< R$, so ω is *larger* than in (a) ✓ — correct, since the rod's mass sits at smaller radius than R and so needs a faster spin to hold the lean.

↑ Two lines. The first proves the correction is consistent with (a); the second shows the sign of the correction was understood rather than stumbled into.

How that gets graded

USAPhO rubrics are not published, so the breakdown below is a reconstruction — but it is the right *shape*: a 25-point problem is cut into five or six milestones, and each is awarded on its own.

Rubric line	Pts	Earned
(a) Point-mass limit		
Correct force picture: states the frame and identifies the centrifugal force acting at the cm	2	2
Correct torque (or force) balance about the contact point	3	3
Correct $\omega = \sqrt{g \tan\theta / R}$	3	3
(b) Extended rod		
Recognizes the centripetal acceleration varies along the rod, so the cm substitution fails	3	3
Correct parametrization $r = R - s\sin\theta$, $z = s\cos\theta$, $dm = (m/h)ds$	4	4
Correct integral expression for the centrifugal torque	4	4
Correct evaluation $m\omega^2 h\cos\theta\left(\frac{R}{2} - \frac{h\sin\theta}{3}\right)$	3	3
Correct gravitational torque and balance	3	3
Correct final ω, boxed	3	3
Total	**25**	**25**

That submission gets full marks, and it is worth noticing how little of the page is polished. There is a crossed-out formula, an abandoned idea, and no prose beyond a handful of six-word fragments. **What makes it a 25 is that every rubric line is findable in under ten seconds.**

The same physics, written four other ways

Here is what actually separates scores on this exam. All four of these students did roughly the same amount of physics.

What the page contains	Score
Ran out of time in (b). Part (a) complete. In (b): "the centripetal acceleration varies along the rod, so I have to integrate the torque instead of using the cm," plus $r = R - s\sin\theta$, $z = s\cos\theta$, $dm = (m/h)ds$, and the integral written down but never evaluated.	**18/25**
Fell into the trap. Part (a) complete. In (b), substituted $R \to R - \frac{h}{2}\sin\theta$ with no comment and boxed it.	**9/25**

What the page contains	Score
Blew part (a), recovered. Part (a) has the lever arms swapped and is never checked, so $\cot\theta$ is boxed. Part (b) is done correctly from scratch — correct setup, integral, evaluation, balance, and answer.	**22/25**
Did all the physics, wrote none of it. Everything correct, but done on scratch paper, with only the two boxed answers copied onto the answer sheets.	**6/25**

Read those four rows again. The third student got *part (a) wrong* and still scored 22, because subparts are graded independently and (b) was written out. The fourth student solved the entire problem correctly and scored 6, because the work was on the wrong paper.
The gap between 6 and 22 on this problem is not physics. It is where you wrote and how you labeled it.

Things that went wrong, and what to do about them

You suspect part (a) is wrong but cannot see why.

Run the limiting cases — that is what caught the swapped lever arms above. Set each parameter to zero or infinity in turn and ask whether the answer still makes sense. If a check fails, you have found a real error in thirty seconds. If they all pass, stop worrying and move on; a wrong (a) does not sink (b) anyway.

You cannot see how to set up the integral in (b).

Write down what you *do* know — "the acceleration varies along the rod, so $\tau_c = \int \omega^2 rz\, dm$" — and then define r, z, and dm one at a time against your diagram. That decomposition is worth 7 of the 17 points in (b) on its own, before you have integrated anything. Half of getting stuck is trying to write the whole setup in one line.

You get an ugly answer and think you must have erred.

Ugly is normal. $\omega^2 = g\tan\theta/(R - \frac{2h}{3}\sin\theta)$ is ugly, and correct. Check limits, box it, and move on. Do not spend ten minutes hunting for a mistake that isn't there because you expected something prettier.

You have two versions of an answer and no time to decide.

Pick the one that passes your limiting checks, box that one, and put a single line through the other. Never leave both live.

You are 30 minutes in with (b) half done.

Stop. Box whatever you have, write one line saying what remains ("still need to evaluate the integral and balance against gravity"), and go read the next problem. Its first subpart is almost certainly worth more than your next ten minutes on this one.

7 Your Checklist

Before the exam

- ☐ Confirmed the date, start time, and site with your proctor
- ☐ Non-graphing scientific calculator, memory cleared
- ☐ Memorized the formulas the reference sheet does *not* give you — it is constants and four approximations, nothing else
- ☐ Practiced on at least two past USAPhOs *under timed conditions*, writing on paper, in 90-minute blocks

During the exam

- ☐ Read all three problems before starting any of them
- ☐ Diagram first, symbols defined on it
- ☐ Name the principle before the algebra
- ☐ Symbols throughout; numbers only at the end
- ☐ Box every subpart's final answer
- ☐ Header on every page: problem number and n/N
- ☐ Solve on the answer sheets, not on scratch
- ☐ Stuck? State an assumed result and do the next part anyway
- ☐ 30 minutes per problem, hard stop
- ☐ Last four minutes: box, label, cross out, sort

Questions this handout does not answer? Ask me — and check aapt.org/physicsteam, which is always the final authority.